

Prasant Koirala

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EDUCATION

The University of Southern Mississippi

Hattiesburg, MS

Bachelor of Science in Computer Science, GPA: 3.94

Expected, May 2027

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Computer Systems

EXPERIENCE

Research Assistant

Sept. 2025 – Present

The University Of Southern Mississippi

Hattiesburg, MS

- Built and trained a ModernBERT-based co-attention neural network in PyTorch for social engineering email detection, achieving 100% accuracy on a held-out test set of 158 email pairs.
- Engineered an email pair processing pipeline from a preprocessed dataset derived from 500,000+ corporate emails, generating 171 structured emails across 4 attacker profiles.
- Trained a 151M+ parameter transformer with AdamW over 20 epochs, reducing validation loss from 0.57 to 0.0009 while maintaining perfect precision, recall, F1, and AUROC.
- Integrated the model with an upstream GNN-based anomaly detector, achieving 92% end-to-end system accuracy for social engineering detection.

Web/App Development Intern

Jun. 2025 – Aug. 2025

Optimal Answers, LLC

Gulfport, MS

- Built and shipped backend features for internal and customer-facing web applications using C#, SQL, and REST APIs, supporting day-to-day workflows for business and operations teams.
- Refactored performance-critical backend logic and optimized database queries, reducing UI latency and cutting employee task time by 30% across core application flows.
- Designed and deployed a staging and pre-release validation environment, enabling safe testing and reducing production incidents by 40%.
- Collaborated closely with design, QA, and data teams to deliver 5+ production features, participating in requirement discussions, testing, and rollout.

PROJECTS

Timeonar | C#, MySQL, TypeScript

[Github](#)

- Built a full-stack web application that converts natural-language queries into interactive literature timelines using a backend LLM/AI API orchestration service.
- Implemented a React + TypeScript frontend and a .NET C# REST API to deliver validated requests and structured, citation-backed responses, improving reliability and clarity of aggregated results.
- Designed backend orchestration workflows implementing multi-stage request pipelines with failure isolation, partial result propagation, and incremental response streaming, reducing perceived latency via server-side execution.

MangOS: x86 Operating System | C, x86, Systems Programming

[GitHub](#)

- Implemented a custom x86 Operating System bootstrapped via the Limine bootloader, transitioning from bootloader state into protected kernel execution with a team of 6 developers.
- Built a kernel-level framebuffer terminal with direct video memory access, text rendering, scrolling, and cursor control without libc.
- Designed low-level memory and I/O abstractions to safely interface with hardware from kernel space.

Personal Status Monitor | C++, Windows, Linux

[GitHub](#)

- Built a cross-platform C++ status monitor that tracks active applications and user status using OS/process inspection with minimal overhead.
- Implemented an authenticated HTTP client (WinHTTP/libcurl) to POST JSON status updates to a portfolio API for real-time display.

TECHNICAL SKILLS

Languages: TypeScript, C++, C, Python, Bash

Systems & Infra: AWS, Docker, Google Cloud, OS internals

Frameworks: FastAPI, Node.js (Express), React, TailwindCSS, GraphQL

Database: PostgreSQL, MongoDB, DynamoDB

AI/ML: Pytorch, TensorFlow, FAISS, Pinecone, LLM APIs